

JFC CNN-TIC / CNN-MIC Pipeline Summary

1. Data Preparation

- Downloaded Jollibee Foods Corporation stock data matching the Ayala Corporation date range.
- Cleaned OHLCV data using JFC_clean.py to standardize columns and date format.
- Ensured date range alignment with macro dataset.
- Macro dataset used: Macro_SMA6_25.xlsx (SMA6–SMA25 precomputed).
- Final aligned date range: 2012-05-14 to 2025-08-19.

2. Technical Indicator Generation

- Generated technical indicators over windows 6–25.
- Total technical features: 15 indicators × 20 windows = 300 columns.
- Output file: JFC_PS_indicators_6_25.csv.

3. Label Generation

- Used centered 11-day turning point labeling (radius=5).
- Rule:
Sell if current close is local maximum.
Buy if current close is local minimum.
Hold otherwise.
- Encoded as Sell=-1, Hold=0, Buy=1.
- Output rows: 3234.

4. Normalization

- Sequential MinMax normalization using train_end=2022-12-30.
- Excluded OHLCV columns from scaling.
- Generated:
JFC_norm_tech_0_1.csv
JFC_norm_combined_0_1.csv
- Saved scaling parameters to JSON for reproducibility.

5. Quantization to uint8

- Converted normalized values (0–1) to 0–255 uint8.
- Technical only: 300 columns.
- Combined dataset corrected to keep macro columns as SMA6–SMA25 only.
- Removed incorrect macro broadcasting that created _Wxx duplicates.

6. CNN Image Construction

- Built image tensors from uint8 CSV.
- Tech-only dataset shape: (3234, 15, 20).
- Tech+Macro dataset shape: (3234, 20, 20).
- Macro rows detected correctly: 5.
- Outputs:
images.npy
labels.npy
index.csv
manifest.json

7. Current Status

Datasets successfully prepared for:

- Baseline CNN-TIC (15x20)
- Proposed CNN-MIC (20x20)

Next logical step:

- Run training script using dataset_dir.
- Evaluate validation blocks.
- Final fit and 2023–2025 test evaluation.